

SMD, SMG, SFG Mixers and flowmakers

60 Hz

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions describe Grundfos SMD, SMG, SFG mixers and flowmakers.

Sections 1-6 give the information necessary to be able to unpack, install and start up the product in a safe way.

Sections 7-11 give important information about the product, as well as information on service, fault finding and disposal of the product.

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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

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1. General information

This booklet includes instructions for installation, operation and maintenance of Grundfos SMD, SMG mixers and SFG flowmakers. These products are designed for applications involving the mixing, i.e. the homogenisation and suspension, of liquids of low to medium viscosity (≤ 500 mPas).

There are separate installation and operating instructions for the AMD.07.18.1410 mixer. See publication number 96526302 on www.grundfos.com.

Note

For information on AMD, AMG and AFG in Ex version, see publication number 97761673 on www.grundfos.com.

1.1 Symbols used in this document



Warning

If these safety instructions are not observed, it may result in personal injury.



Warning

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

Note

Notes or instructions that make the job easier and ensure safe operation.

2. Safety instructions

2.1 General safety instructions



Warning

Before starting any work on mixers or flowmakers, make sure that the fuses have been removed or the mains switch has been switched off. Make sure that the power supply cannot be accidentally switched on.

Warning

Follow these safety instructions as well as the instructions in each individual section when transporting, storing, handling and operating the mixer or flowmaker.



The mixer or flowmaker must be installed, connected, started up and serviced by qualified persons.

Beware of rotating parts.

Make sure that persons cannot accidentally fall into the tank, e.g. by installing a cover or railing.

3. Receiving the product

3.1 Transporting the product

The individual components of the mixer or flowmaker must be packed carefully to prevent any damage to the surface protection during transportation.



Warning

Make sure that the mixer or flowmaker cannot roll or fall over.

Warning



Observe any local regulations that set limits for the weight of the components to be lifted manually by individuals, i.e. handled without the use of lifting equipment.

Make sure that all lifting equipment is rated for the purpose. Check the lifting equipment for damage before any attempts to lift the components are made. Do not exceed the lifting equipment rating.

3.2 Inspecting the product



Warning

Do not install a damaged mixer or flowmaker.

On delivery, check the mixer or flowmaker and any accessories supplied with it for transport damage. This also applies when the equipment is delivered to the installation site.

If the mixer, flowmaker or any accessories have been damaged, contact your local Grundfos company before continuing to install the equipment. Do not remove a damaged new component for further inspection, unless instructed by your local Grundfos company.

Dispose of packaging material according to local regulations.

3.3 Storing the product

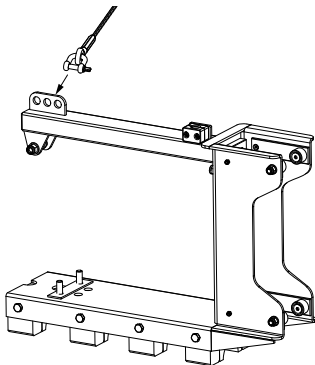
Store mixers or flowmakers in a dry location in which the temperature is not subject to major fluctuations.

Do not expose the flowmaker propeller blades to direct sunlight for more than one month.

If the mixer or flowmaker has been stored for more than one year, change the gearbox oil. Change the oil even if the unit has never been in use. This is necessary because of natural aging of mineral oil lubricants.

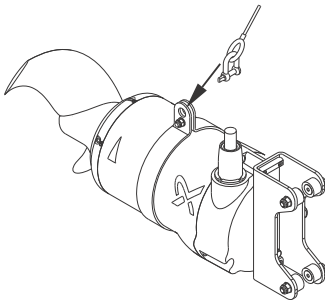
4. Installing the product

Only lift the mixer or flowmaker using the suspension point. See figures 1 and 2.



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Fig. 1 Mounting of lifting wire on motor bracket



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Fig. 2 Mounting of lifting wire on SMD

Note Use the lifting equipment supplied with the unit only for lifting the mixer or flowmaker.

Caution Never hang the mixer or flowmaker by the power supply cable. Never let the mixer or flowmaker run while hanging in the lifting equipment.

See pages 31-33 for details about weights.

4.1 Torques

All nuts and screws used for the installation must be made of stainless steel.

Use grease, alu-paste, together with a spring washer or locking nut. Otherwise, use Loctite or a similar product for lubrication and locking.

Tighten all nuts and screws to the following torques:

	Screws F-class 70 [Nm]	Screws F-class 80 [Nm]
M6	8.8	11.8
M8	21.4	28.7
M10	44	58
M12	74	100
M16	183	245
M20	370	494

4.1.1 Anchor bolts

Anchor bolts used for mounting components in concrete must have the following pull-out strength:

Size	Pull-out strength [kN]
M12	6
M16	14

4.2 Positioning of mixers and flowmakers

Correct positioning of mixers and flowmakers is essential to ensure trouble-free operation and long life.

4.2.1 Mixers

Submerge the mixer as deeply as possible.

- Place the mixer in such a way that you obtain a good mixing of the liquid in the entire tank. If more mixers are installed in the same tank, the mixers must not generate opposite flows.
- The distance between the propeller tip and the bottom of the tank, H_{MIN} in fig. 3, must be half of the propeller diameter.
- SMG:** The distance from the propeller tip to the liquid surface, H_{ABOVE} in fig. 3, must be at least equal to the propeller diameter.
- SMD:** The distance from the propeller tip to the liquid surface, H_{ABOVE} in fig. 3 must be at least 1.5 times the propeller diameter.
- The distance between the propeller tip and the wall behind the mixer, L_{MIN} in fig. 3, must be at least 1.5 times the propeller diameter.

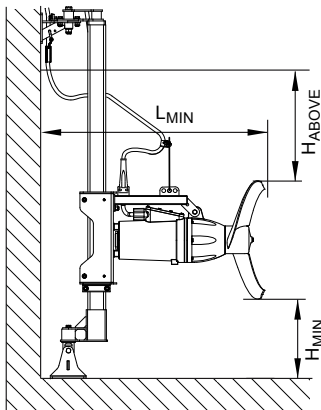


Fig. 3 Positioning sketch for mixers

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4.2.2 Flowmakers

Submerge the flowmaker as deeply as possible.

- The distance between the propeller tip and the bottom of the tank, H_{MIN} in fig. 4, must be 50 cm.
- The distance from the propeller tip to the liquid surface, H_{ABOVE} in fig. 4, must be at least 0.75 times the propeller diameter.
- The distance between the propeller tip and the wall behind the flowmaker, L_{MIN} in fig. 4, must be at least twice the propeller diameter.
- The sideways distance between the propeller tip and the tank wall must be at least 0.5 metres.
- If more flowmakers are to be installed in parallel, the distance between their propellers must be larger than half the propeller diameter.
- The distance from a bend in the channel to the propeller and from the propeller to aeration areas must be at least the larger value of channel width and water level.

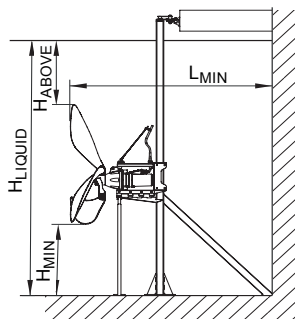


Fig. 4 Positioning sketch for flowmakers

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4.3 Installing a mixer

You can mount the mixer in different ways depending on the accessories:

- fixation bracket and tube for suspended mounting. See fig. 5
- fixation bracket for wall mounting. See fig. 5
- fixation base for floor mounting. See fig. 5
- motor bracket for mounting on column profile tube. See fig. 6.

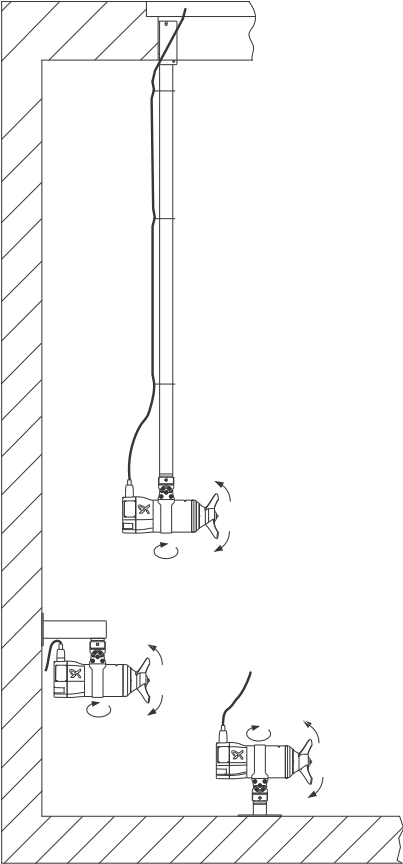


Fig. 5 SMD.09 - 18.xx.xxxx.T

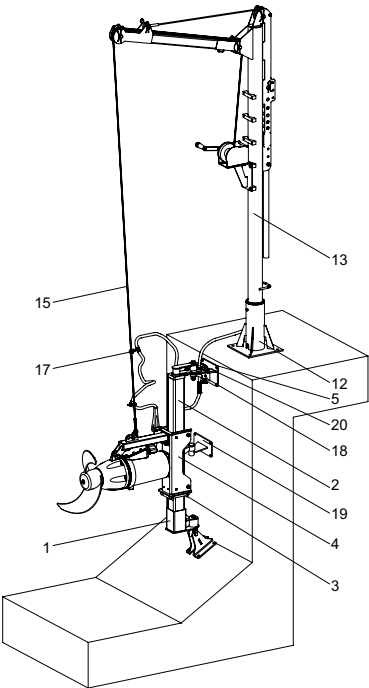


Fig. 6 Installation of SMD and SMG mixers

Position numbers in figures 6 to 10

Pos.	Description
1	Bottom fixation bracket
2	Column profile
3	Depth blocker
4	Motor bracket
5	Top fixation bracket incl. safety wire
12	Crane foot
13	Crane with winch
15	Lifting wire incl. shackle and wire clamp
17	Cable clamp
18	Cable sock incl. shackle
19	Intermediate fixation bracket
20	Wire clamp
31	Hole for fastening of safety wire

Procedure

See fig. 6

Torques appear from section 4.1 *Torques*.

1. Drill the holes for the mounting screws for the top fixation bracket.
2. Mount the screws, and fit the top fixation bracket.
3. Place the bottom fixation bracket in the right position, vertically below the top fixation. You can mount the bottom fixation bracket in any angle from vertical to horizontal position.
4. Drill the holes for the mounting screws for the bottom fixation bracket.
5. Mount the screws, and fit the bottom fixation bracket.
6. Depending on the length of the column profile, weld the turnable part of an intermediate fixation bracket (19, fig. 6) to the column profile.
7. Position and align the column profile in the bottom fixation bracket. Shorten the column profile (2) to the correct length to match the position of the insulator (C) at the top fixation bracket. A gap of 5 to 10 mm between collar of insulator and column profile is optimal. See fig. 7.

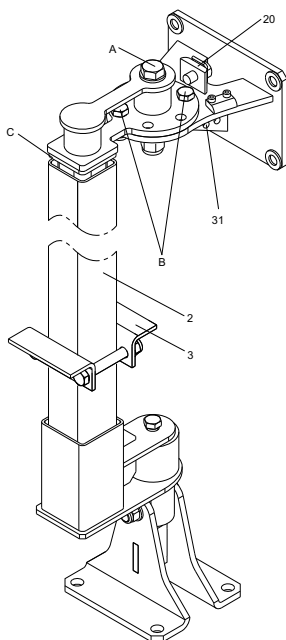


Fig. 7 Fixation

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8. Remove the insulator and the turnable metal part by removing the centre screw (A) and the two fixation screws (B).
9. Adapt the outside of the square insulator to the inside of the column profile. The insulator must fit tightly inside the column profile.
10. Insert the column profile into the bottom fixation bracket, and mount the top end with the insulator and the turnable metal part on the already installed top fixation bracket. Tighten the three screws (A and B) in the desired position. You can adjust the angle in steps of 7.5 °.
11. If you use an intermediate fixation bracket (19), fit it to the turnable part welded on to the column profile in step 6. Drill holes in the tank wall, fit screws in the bracket and tighten the screws.

Caution Make sure that the mixer cannot be turned so much that the propeller touches the tank wall.

12. Fasten the depth blocker (3) in the right position.
13. Drill the holes for the mounting screws for the crane foot (12, fig. 6).
14. Mount the crane foot, and fit and tighten the screws.
15. Mount the lifting wire (15) on the motor bracket using the shackle. See fig. 9.
16. Mount the top end of the safety wire to the hole (31) of the top fixation bracket using a shackle. The other end of the safety wire ends in a shackle through which the lifting wire must run.

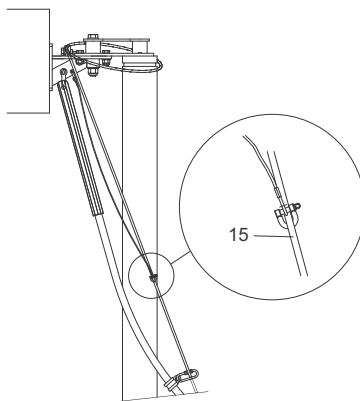


Fig. 8 Safety wire

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17. Fix the power supply cable to the lifting wire by means of a cable clamp approx. 0.8 m above the mixer. This will prevent the cable from falling down and becoming entangled in the propeller during operation. Connect the cable clamp to the lifting wire above the wire clamp by means of a snap hook. See fig. 9. Attach the power supply cable to the lifting wire by means of cable clamps placed at 1 m intervals.

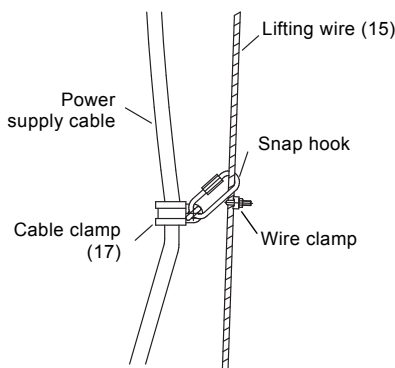


Fig. 9 Attaching the power supply cable to the lifting wire

18. Position the crane in the foot, and mount the lifting wire in the drum of the winch.

Caution At least three turns of wire must always remain on the drum. Otherwise the wire may break loose from the drum fixation.

Note Follow the separate installation and operating instructions for cranes.

19. Lift the complete mixer, motor bracket with motor, using the crane and slide it over the column profile.
20. Slowly lower the mixer into the tank and down to its position on the depth blocker.
21. Mount the cable sock (18) to the top fixation bracket using the shackle, and pull the motor cable through it to the desired position. See fig. 10. The power supply cable must be slightly tightened.

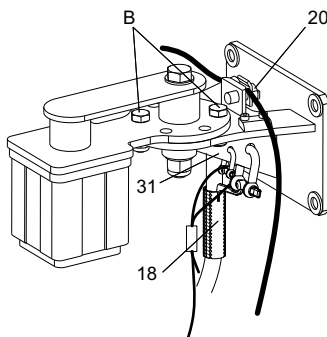


Fig. 10 Top fixation bracket with lifting and safety wires and cable sock

Caution Remove the lifting wire from the crane before starting the mixer.

22. Remove the lifting wire from the winch and fit it to the wire clamp (20, fig. 10) on the top fixation bracket.

Use the lifting wire as a relief for the power supply cable. For this reason, always tighten the lifting wire.

23. Check the distance between the propeller and the wall and the propeller and the bottom of the tank when the motor bracket is resting on the depth blocker. The mixer must under no circumstances touch other installations, bottom or wall. This also applies when the mixer is turned. See section [4.2 Positioning of mixers and flowmakers](#).



Warning

Make sure that the power supply to the control cabinet has been switched off.

24. Connect the power supply cable to the terminals in the control cabinet.

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4.4 Installing a flowmaker

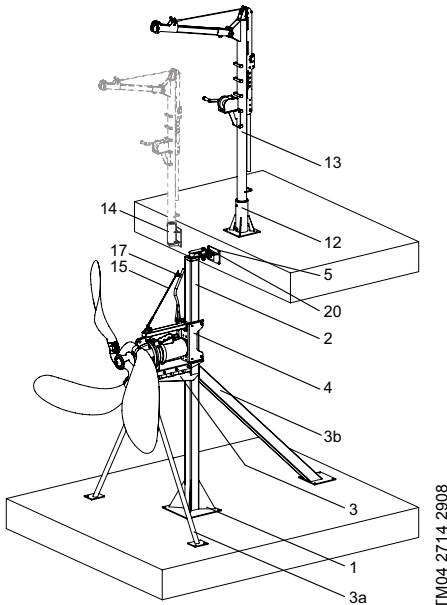


Fig. 11 SFG flowmaker

Position numbers in figures 11 and 19

Pos.	Description
1	Bottom fixation plate
2	Column profile
3	Depth blocker
3a	Front support leg
3b	Back support leg
4	Motor bracket
5	Top fixation bracket incl. safety wire
12	Crane foot
13	Crane with winch
14	Crane foot for vertical installation
15	Lifting wire incl. shackle and wire clamp
17	Cable clamp
20	Wire clamp
31	Hole for fastening of safety wire

Procedure

Torques appear from section 4.1 [Torques](#).

Note

You can mount the propeller blades before installation of SFG.xx.180/230/260 or after you have installed the crane.

4.4.1 Mounting the propeller blades of SFG.xx.180/230

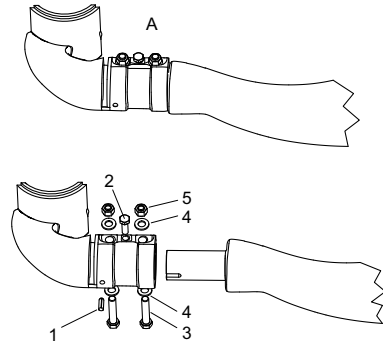


Fig. 12 SFG.xx.180/230

The fixation of the propeller blades is ensured by the pressure of the clamping jaws formed by the screws (3) and nuts (5). The function of the pin (1) is to position the blades in the right angle before tightening the clamping jaws.

Caution

1. Check the premounted pins (1). Only for positioning.
2. Tighten the screw (2) to widen the opening of the clamping jaws.
3. Insert the blade from above. Turn it slightly and let it rest on the pin. Turn the blade back until the blade slides down and is flush with the hub.
4. Remove the screw (2).
5. Put Loctite 243 threadlocker on the thread of the screw (3).
6. Mount position 3, 4 and 5 in both holes and tighten by hand.
7. Tighten both nuts (5) with a torque wrench to 100 Nm (A4-80).
8. Check visually that the blade actually looks like position A in fig. 12.
9. Apply silicone to the joint between blade and hub.

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4.4.2 Mounting the propeller blades of SFG.xx.260

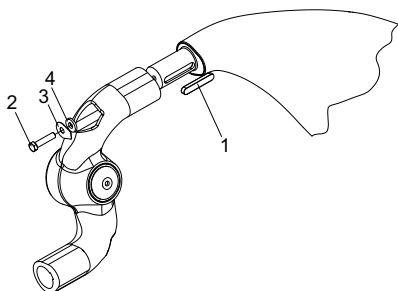


Fig. 13 SFG.xx.260

1. Drive the key (1) into the keyway for correct positioning of the blades.
2. Apply a little oil to the blade shaft and the hole in the hub.
3. First put the cover plate (3) and then the washer (4) on the screw (2).
4. Put Loctite 243 threadlocker on the thread of the screw (2).
5. Turn the hub, and insert the blade from above.
6. Turn the blade to correct position and let it slide into the hub.
7. Screw in the screw (2) by hand and tighten it with a torque wrench to 183 Nm (A2-70).
8. Carry out a visual check.
9. Apply silicone to the joint between the blade and the hub and between the cover plate (3) and the hub.

4.4.3 All flowmakers

1. Weld the bottom fixation plate on the end of column profile in a workshop.

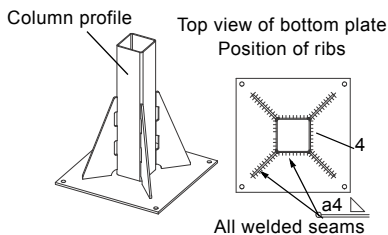


Fig. 14 Welding the column profile on to the bottom fixation plate

Note The SFG.xx.130 bottom fixation plate does not have ribs.

2. SFG.xx.260:

Place the depth blocker (3), the back support leg (pos. 3b) and the front support legs (3a) in the right position, and weld them on the column profile (2). See position numbers in fig. 16.

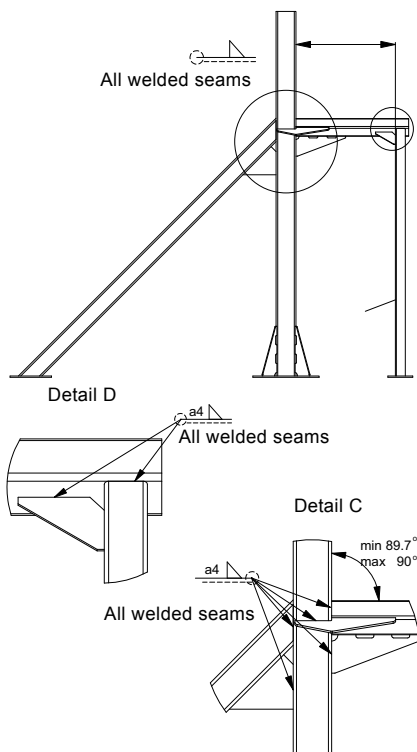
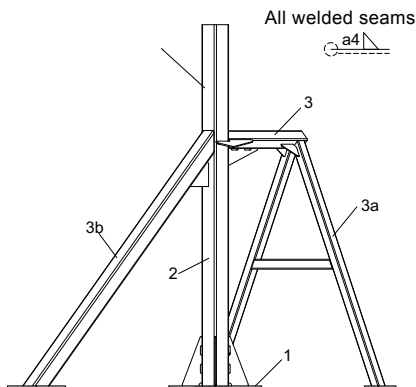


Fig. 15 SFG.xx.260, profile and support legs, side view



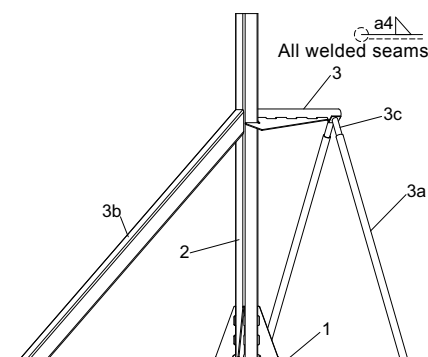
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Fig. 16 SFG.xx.260, profile and support legs, back view

3. SFG.xx.180 and SFG.xx.230:

Place the depth blocker (3), the back support leg (3b) and the front support legs (3a and 3c) in the right position, and weld them to the column profile (2) in a workshop. See position numbers in fig. 17.

Calculate the required height of the SFG depth blocker according to the positioning rules for flowmakers. See section 4.2.2 *Flowmakers*.



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Fig. 17 SFG.xx.180-230 profile and support legs

Note

There must be a minimum overlap of 150 mm between the parts of front support legs (3a and 3c). See fig. 18.

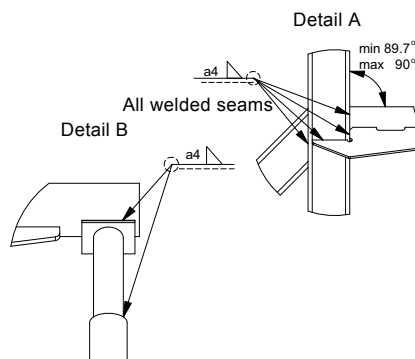
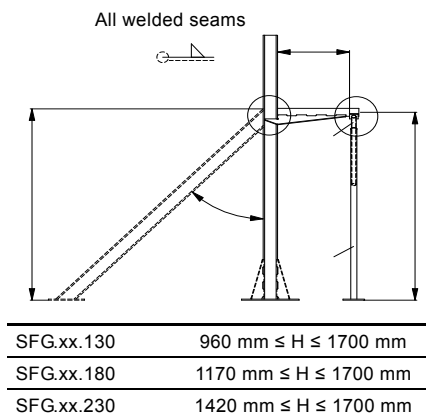


Fig. 18 SFG.xx.130-230 profile and support legs

4. SFG.xx.130:

Place the depth blocker (3) and the front support legs (3a and 3c) in the right position, and weld them to the column profile (2) in a workshop. See position numbers in fig. 17.

Calculate the required height of the SFG depth blocker according to the positioning rules for flowmakers. See section 4.2.2 *Flowmakers*.

Note

The back support leg (3b) is not supplied for SFG.xx.130.

Note

There must be a minimum overlap of 150 mm between the parts of front support legs (3a and 3c). See fig. 18.

- Drill the holes for the mounting screws for the top fixation bracket.

6. Mount the screws, and fit the top fixation bracket.

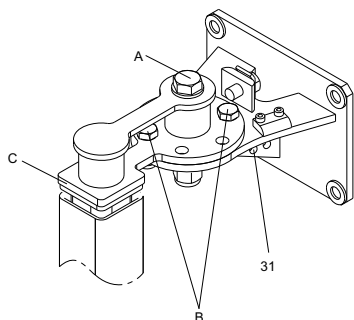


Fig. 19 Top fixation

7. Shorten the column profile (5, fig. 16) to the correct length to match the position of insulator (C, fig. 19) at the top fixation bracket. A gap of 5 to 10 mm between collar of insulator and column profile is optimal. See fig. 19.
8. Remove the insulator and the turnable metal part from the top fixation bracket by removing the centre screw (A) and the two fixation screws (B). See fig. 19.
9. Adapt the outside of the square insulator to the inside of the column profile. The insulator must fit tightly inside the column profile.
10. Fit the turnable metal part, now fitted on top of the column profile, to the already mounted top fixation bracket. Tighten the three screws (A and B in fig. 19) in the desired position. You can adjust the angle in steps of 7.5° .
11. Drill the holes for the mounting screws for the bottom fixation plates in the bottom of the tank, and insert the screws. See fig. 16.
12. Tighten the screws in the bottom fixation plate.
13. Drill holes, mount and tighten the screws of the front and back support legs as well.
14. Drill the holes for the mounting screws for the crane foot in the concrete.
15. Mount the crane foot, and fit and tighten the screws.

16. Mount the lifting wire (15) on the motor bracket using the shackle. See fig. 20.

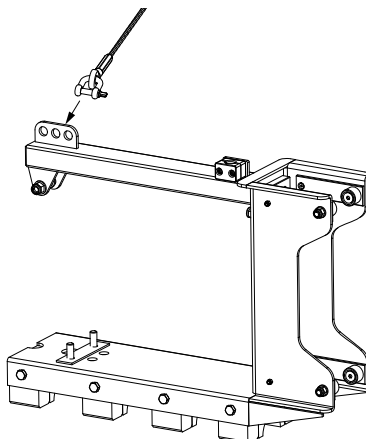


Fig. 20 Mounting of the lifting wire on the motor bracket

17. Mount the top end of the safety wire to the hole (31 in fig. 19) of the top fixation bracket using a shackle. The other end of the safety wire ends in a shackle through which the lifting wire must run.

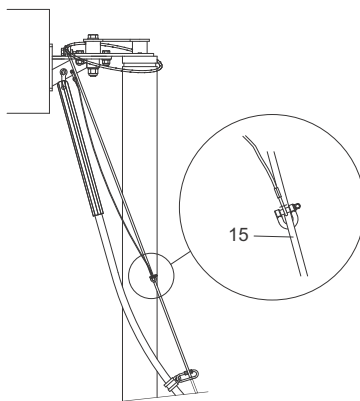


Fig. 21 Safety wire

18. Fix the power supply cable to the lifting wire by means of a cable clamp approx. 0.8 m above the flowmaker. This will prevent the cable from falling down and becoming entangled in the propeller during operation. Connect the cable clamp to the lifting wire above the wire clamp by means of a snap hook. See fig. 22. Attach the power supply cable to the lifting wire by means of cable clamps placed at 1 m intervals.

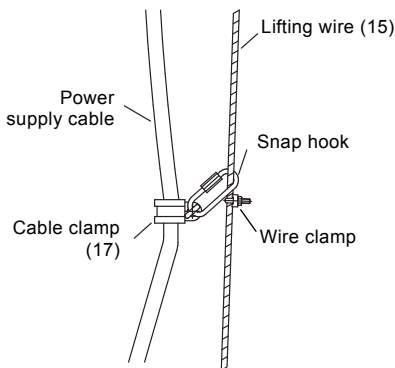


Fig. 22 Attaching the cable to the wire

19. Position the crane in the foot, and mount the lifting wire in the drum of the winch.

Caution

At least three turns of wire must always remain on the drum. Otherwise the wire may break loose from the drum fixation.

Note

Follow the separate installation and operating instructions for cranes.

20. Lift the complete flowmaker, motor bracket with motor, using the crane and slide it over the column profile.
21. Slowly lower the flowmaker into the tank and down to position on the depth blocker.

22. Mount the cable sock (18, fig. 23) to the top fixation bracket using the shackle, and pull the motor cable through it, to the desired position. See fig. 23. The power supply cable must be slightly tightened.

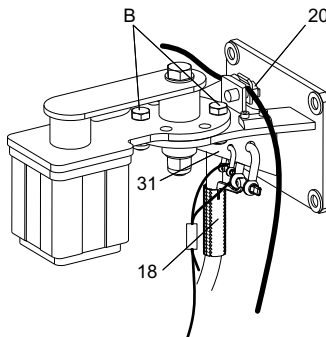


Fig. 23 Top fixation bracket with lifting and safety wires and cable sock

Caution

Remove the lifting wire from the crane before starting the flowmaker.

23. Remove the lifting wire from the winch and fit it to the wire clamp (20) on the top fixation bracket.
24. Use the lifting wire as a relief for the power supply cable. For this reason, always tighten the lifting wire. See fig. 22.
25. Check the distance between the propeller and the wall and the propeller and the bottom of the tank when the motor bracket is resting on the depth blocker. The flowmaker must under no circumstances touch other installations, bottom or wall.



Warning

Make sure that the power supply to the control cabinet has been switched off.

26. Connect the power supply cable to the terminals in the control cabinet.

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5. Electrical connection

Electrical connections must be carried out by a qualified electrician in accordance with local regulations.

Observe all national and local regulations relating to safety and accident prevention.



Warning

Before making any electrical connections, make sure to remove the fuses or switch off the mains switch. Make sure that the power supply cannot be accidentally switched on.

The supply voltage and frequency are marked on the mixer or flowmaker nameplate. Make sure that the mixer or flowmaker is suitable for the power supply available at the installation site.

The mixer or flowmaker is supplied complete with a power supply cable of 10 m. Standard length, suitable for up to 7 m deep tanks. Standard cable lengths are 10 and 15 m, longer cables are available on request. See section [10.1 General technical data](#).

The motor is marked either with a Y, star, or a Δ, delta. For 10- and 11-wire cables, this connection is to be made in an external control panel using conductors 1 to 6 of the power supply cable.

Figure 24 shows a schematic drawing of these star and delta connections. See also section [5.5 Wiring diagrams for SMG and SFG](#).

If the mixer or flowmaker is connected in delta during operation, you can start it in a star-delta connection.

Three-phase motors

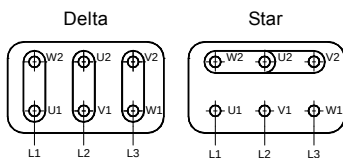


Fig. 24 Schematic drawing of delta and star connection

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5.1 Motor protection

Mixers and flowmakers are provided with the following types of motor protection:

Standard mixers incorporate three bimetallic PTO thermal switches. See fig. [27](#).

Standard flowmakers incorporate three PTC sensors according to DIN 44081. See fig. [28](#).

Function of thermal switches

The motor is protected against overheating by three thermal switches connected in series, one switch in each winding.

When the maximum winding temperature is reached, the switch will open the circuit and stop the motor.

When the windings have cooled to normal temperature, the switch closes the circuit and you can restart the motor. Manual restarting is necessary.

See fig. [27](#), section [5.5 Wiring diagrams for SMG and SFG](#).

Thermal switches (F6)

- Two conductors, terminals 11 and 12.
- Maximum operating voltage: 250 V.
- Maximum switching current: 2.5 A at $\cos \varphi = 1$.
- Cutting-out temperature: 150 °C.

Function of PTC sensors

When overheated, the motor stops. Automatic restarting is not permitted in such cases. This requires a thermistor trigger unit with a reconnection suppressor in the control circuit of the motor contactor.

See wiring diagram in fig. [28](#), section [5.5 Wiring diagrams for SMG and SFG](#).

Ø1, Ø2, Ø3: PTC sensors:

- Two conductors, terminals 31 and 32.
- Maximum voltage at the terminals:
 $U_{\max.} = 2.5 \text{ V (AC/DC)}$.
- Resistance between terminals 31 and 32:
 - at room temperature $R = 150 \text{ to } 750 \ \Omega$
 - at cutting-out temperature (130 °C)
 $R \geq 4000 \ \Omega$.

Note

When you have tested the connection at terminals 31 and 32, the test voltage must not exceed 2.5 V (AC/DC). Use an ohmmeter for the test.

5.2 Gearbox/shaft seal housing protection

The gearbox/shaft seal housing is monitored for the ingress of water by means of a leakage sensor incorporated in the gearbox/shaft seal housing.

If the monitoring function is required, connect the leakage sensor to a Grundfos relay, type ALR-20/A. Order the relay separately. Part number: 96489569.

If the gearbox/shaft seal housing is not monitored for the ingress of water, we recommend that you make inspections every 6 months. If the oil contains water, replace the shaft seal.

Note

The cable between the relay and the mixer or flowmaker must not be longer than 50 m.

For a longer distance, use an additional, screened cable. Always connect an external alarm indicator, if fitted, to the potential-free outputs, terminals 1 and 3 or 4. Maximum load: 250 V, 5 A.

When the ALR-20/A relay is connected, a current of up to 10 mA flows through the leakage sensor. Terminals 5 and 7 in fig. 25 connected to wires 21 and 22 in fig. 27. In case water penetrates into the oil chamber, the relay will trigger an alarm signal and/or switch off the motor.

See fig. 27 or 28, section 5.5 *Wiring diagrams for SMG and SFG*.

Leakage sensor

- Two conductors, wires 21 and 22.
- Maximum operating voltage: Approx. 12 V.
- Maximum current: 1 to 10 mA.



Warning

When adjusting the relay, beware of electric voltage.

To adjust the sensitivity of the ALR-20/A relay, proceed as follows. See fig. 25.

1. Turn the adjusting screw on the relay (a) until the indicator light (b) lights up.
2. Turn the adjusting screw in the opposite direction until the indicator light turns off.
3. Continue to turn the adjusting screw 60 °. Same direction as under point 2.

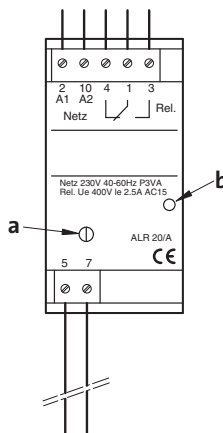


Fig. 25 ALR 20/A relay



Fig. 26 ALR 20 relay connections

Note

Do not check the leakage sensor with an ohmmeter or other measuring instruments. The leakage sensor is an electronic component.

5.3 Overload relays

Protect the motor must against overload via a thermal delay relay according to local regulations. Adjust the relay to the rated current stated on the nameplate.

In the case of star-delta starting, the adjustable value is to be rated 0.58.

Incorporate electro-thermal all-pole triggers in all six mains conductors (U1, V1, W1 and U2, V2, W2).

TM02 8866 0904

TM03 2060 3505

5.4 Starting method

5.4.1 SMD

Continuous operation

Direct start is possible throughout the entire power range.

Intermittent operation

For motors of 1.9 kW and higher, we recommend that you use a soft starter or frequency converter.

5.4.2 SMG

Continuous operation

You can start motors up to 1.6 kW via direct starting. We recommend star-delta starting, soft starter or frequency converter for motors of 2.0 kW and higher.

Intermittent operation

We recommend star-delta starting, soft starter or frequency converter throughout the entire power range.

5.4.3 SFG

Start flowmakers via a star-delta starting, soft starter or frequency converter.

5.5 Wiring diagrams for SMG and SFG

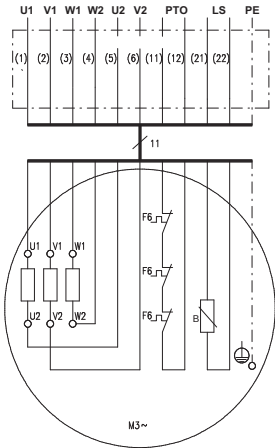


Fig. 27 Three thermal switches

Terminals	Description
1, 2, 3, 4, 5, 6	Ends of the three stator windings (U1, U2, V1, V2, W1, W2)
11, 12	Thermal switches (F6)
21, 22	Leakage sensor in gearbox (B)

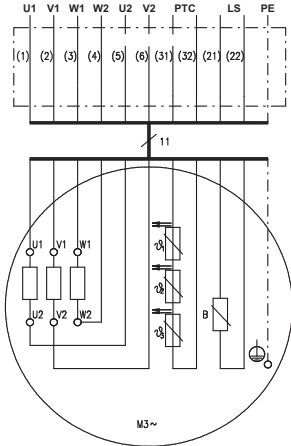


Fig. 28 Three PTC sensors

Terminals	Description
1, 2, 3, 4, 5, 6	Ends of the three stator windings, U1, U2, V1, V2, W1, W2
31, 32	PTC sensors according to DIN 44081 (S1, S2, S3)
21, 22	Leakage sensor in gearbox (B)

TM02 4932 2002

TM02 4940 1802

5.6 Wiring diagrams

Marking		Switch and sensor			Max. load	Connection Relay
Wire 4 / (7*)	Wire 6 / (9*)	Thermal switch	Moisture switch (M)	Leakage sensor (LS)		
11	12	PTO	No	No	2.5 A (250 V)	-
31	32	PTC	No	No	2.5 V	Thermistor
11	13	PTO	Yes	No	2.5 A (250 V)	-
31	33	PTC	Yes	No	2.5 V	Thermistor
11	23	PTO	Yes	Yes	12 V - 11 mA	ALR20/A

* 10-wire cables.

7-wire connections

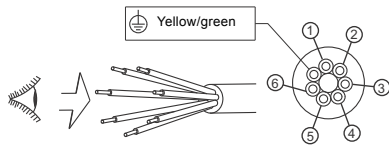


Fig. 29 SMD, 7-wire cable

TM06 5367 4415

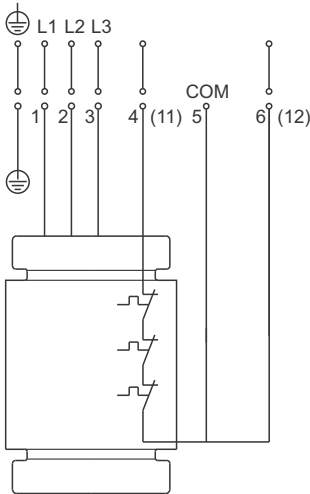


Fig. 30 SMD, 7-wire cable, PTO, standard version

TM06 5362 4415

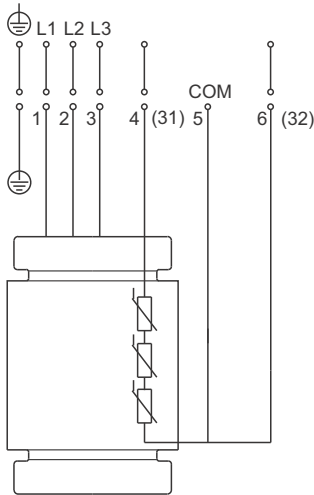


Fig. 31 SMD, 7-wire cable, PTC version

TM06 5363 4415

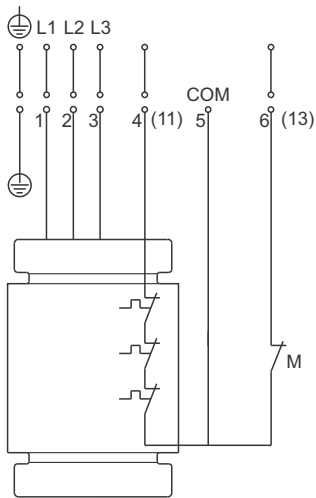


Fig. 32 SMD, 7-wire cable, PTO, moisture switch version

TM06 5364 4415

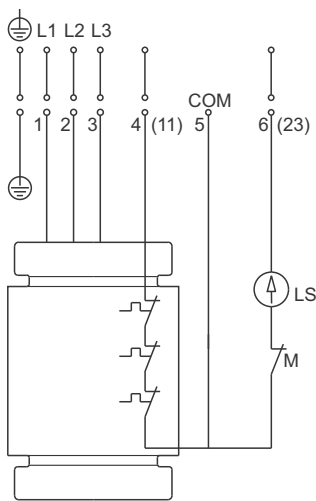


Fig. 34 SMD, 7-wire cable, PTO, moisture switch and leakage sensor version

TM06 5366 4415

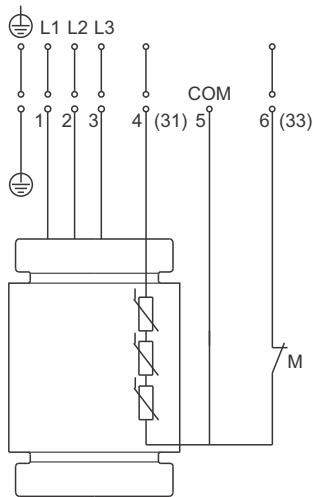


Fig. 33 SMD, 7-wire cable, PTC, moisture switch version

TM06 5365 4415

10-wire connections

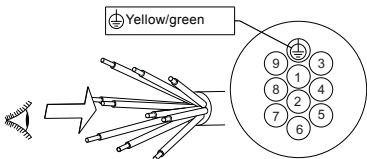


Fig. 35 SMD, 10-wire cable

TM06 5371 4415

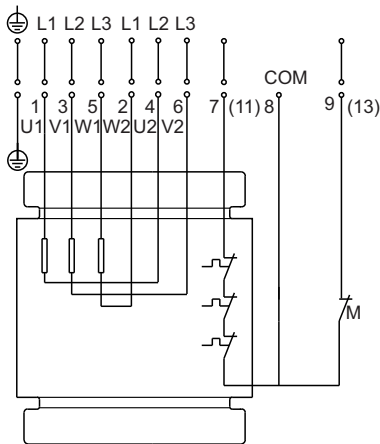


Fig. 36 SMD, 10-wire cable, PTO, moisture switch, standard version

TM06 5368 4415

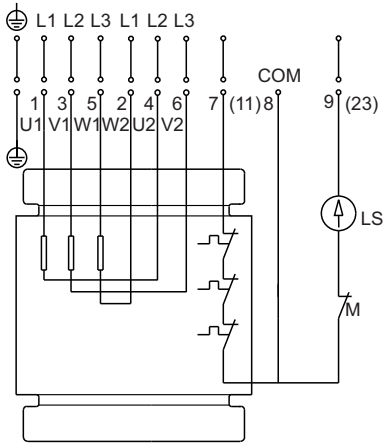


Fig. 38 SMD, 10-wire cable, PTO, moisture switch and leakage sensor version

TM06 5370 4415

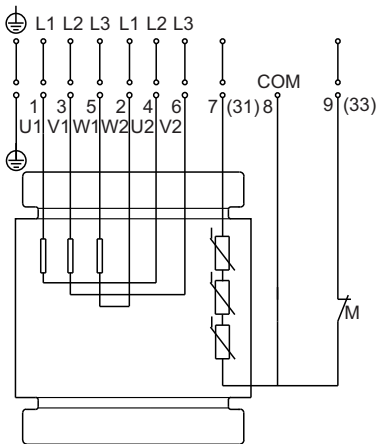


Fig. 37 SMD, 10-wire cable, PTC, moisture switch version

TM06 5369 4415

5.7 Direction of rotation

When you have carried out the electrical connections, check that the propeller is rotating in the correct direction. When viewed from the motor, the propeller must rotate clockwise. An arrow on the motor housing shows the correct direction of rotation.

If the propeller rotates in the wrong direction, interchange two phases of the mains supply.

5.8 Protection against electro-chemical corrosion

Two different metals or alloys cause electro-chemical corrosion if they are connected by an electrolyte. This applies if you install more than one mixer or flowmaker in the same tank. We recommend one of the following additional protection methods:

- galvanic separation of the earth conductor from the neutral conductor
- galvanic separation of the mains supply by means of isolation transformer
- anode kit.

Separate the earth conductor in such a way that no direct current can flow through it. It must still function as a protective conductor. You achieve this with a limiting unit, polarisation cell, anti-parallel diode or an isolation transformer.

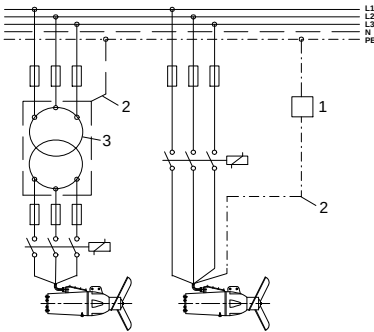


Fig. 39 Protection against electro-chemical corrosion

Pos.	Description
1	Limiting unit
2	Earth conductor
3	Isolation transformer



Warning

When using an isolation transformer, do not alter the ratio between starting current and rated current.

6. Starting up the product

Check the oil level in the gearbox/shaft seal housing. The oil must fill up between 50 and 75 % of the gearbox/shaft seal housing.



Warning

As pressure may have built up in the oil chamber, loosen the oil screw slowly. Do not remove the screw until the pressure has been fully relieved.

If required, fill oil into the gearbox/shaft seal housing through the oil-filling hole (2). See figures 44 and 45. For oil quality and quantity, see section 8.3.1 *Oil quality, gearbox/shaft seal housing*.

If the mixer or flowmaker has been stored and not used for a long period before startup, see section 8.2 *Service chart*.

Checks before startup:

1. Check that the propeller is rotating in the correct direction. See section 5.7 *Direction of rotation*.
2. Make sure that the mixer/flowmaker is completely submerged in the liquid.

Caution

The mixer/flowmaker must always be submerged during operation.

3. Make sure that there are no solid objects in the tank.



Warning

Make sure that no persons can fall into the tank.

TM02 4943 0603

7. Product introduction

7.1 Product description

The mixers are fitted with motors of 0.9 to 18.5 kW.
The flowmakers are fitted with motors of 0.7 to 8.0 kW.

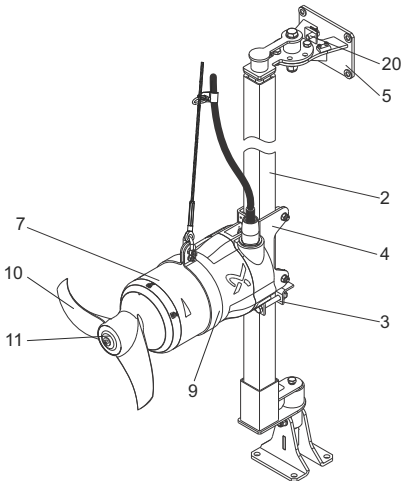


Fig. 40 SMD mixer

TM06 5347 4415

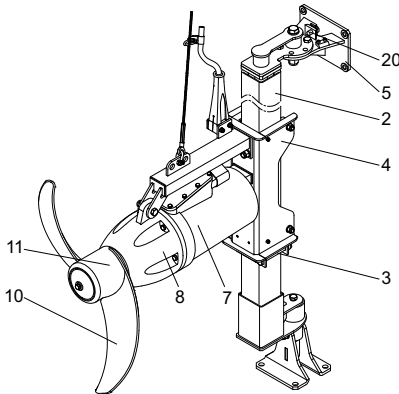


Fig. 41 SMG mixer

TM04 2710 2713

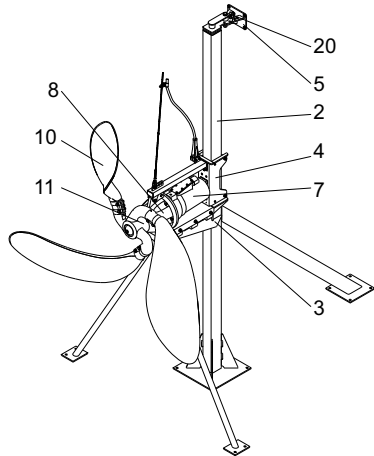


Fig. 42 SFG flowmaker

TM04 2755 2713

Pos.	Description
2	Column profile
3	Depth blocker
4	Motor bracket
7	Motor housing
8	Gearbox
9	Clamping ring
10	Propeller
11	Hub

7.2 Applications

Grundfos mixers and flowmakers are designed for mixing applications within these areas:

- municipal and industrial wastewater treatment
- industrial processes
- sludge treatment
- agriculture.

In order not to overload the mixers and flowmakers and expose them to corrosion, observe the following liquid limitations

Liquid temperature	5-40 °C (60 °C)
pH value	4-10
Maximum dynamic viscosity	500 mPas
Maximum density	1060 kg/m ³
Chloride content	≤ 200 mg/l for stainless steel 1.4301
Chloride content	≤ 1000 mg/l for stainless steel 1.4404

* The liquid temperature range is stated on the nameplate. See section [4. Installing the product](#).

For the mixing of liquids exceeding the values stated above, contact Grundfos.

7.2.1 Mixers

Mixers are suitable for applications involving sludge with a typical dry solids content as stated in the table below. Mixers are also suitable for a wide range of other applications involving similar liquids such as slurry and paper pulp.

Activated sludge	0.5 % DS
Selector zones	0.5 % DS
Anoxic zones	0.5 % DS
Bivalent zones	0.5 % DS
Anaerobic zones	0.5 % DS
Primary sludge	≤ 3 % DS
Secondary sludge	≤ 6 % DS
Digested sludge	≤ 8 % DS
Wastewater tank without screen	≤ 2 % DS
Wastewater tank with sand	≤ 2 % DS

7.2.2 Flowmakers

Flowmakers are suitable for activated sludge with a typical dry solids content of 0.5 to 1.0 % and for other liquids with a dry solids content of maximum 1.5 %.

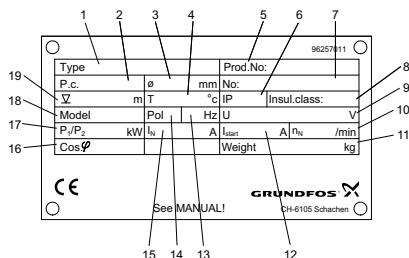
7.3 Identification

7.3.1 Type key

Code	Example	S	M	G.	30.	45.	325.	5.	0B.
S	Type range SMD, SMG, SFG								
M	Version Mixer								
F	Flowmaker								
G	Drive Gear-driven								
D	Direct-driven								
30	Motor output power P2 Code from type designation / 10 [kW] 3.0 kW								
45	Propeller diameter [cm] Code from type designation x 10 [cm] 45 cm								
[.]	Application method Standard								
M	Mud, for higher density								
H	Heavy-duty, biogas plants								
325	Propeller speed [min⁻¹] 325 RPM								
[.]	Installation method Standard								
T	2" thread connection								
[.]	Explosion protection Non-explosion-proof								
Ex	Explosion-proof								
5	Frequency 50 Hz								
6	60 Hz								
0B	Supply voltage 3 x 400-415 V, Y								
1B	3 x 400-415 V, D								
0K	3 x 380 V, Y								
1K	3 x 380 V, D								
0P	3 x 440-480 V, Y								
1P	3 x 440-480 V, D								
[.]	Generation First generation								
A	Second generation								
B	Third generation								
Z	Custom-built products								

7.3.2 Nameplate

You can identify all mixers and flowmakers by means of the nameplate on the motor housing. The details on the nameplate are required for ordering of spare parts.



TM06 2588 4414

Fig. 43 Nameplate

Pos.	Description
1	Type designation
2	Production code
3	Propeller diameter
4	Liquid temperature
5	Product number
6	Enclosure class according to IEC
7	Serial number
8	Insulation class
9	Rated voltage
10	Rated speed, propeller
11	Weight
12	Starting current
13	Frequency
14	Number of poles
15	Rated current
16	Power factor
17	Motor power, P1/P2
18	Model
19	Maximum installation depth

The additional nameplate supplied must be fixed in a visible position at the installation site.

8. Service

Warning



Before starting any work on mixers or flowmakers, remove the fuses or switch off the mains supply. Make sure that the power supply cannot be accidentally switched on.

All rotating parts must have stopped moving.

Before starting any work on mixers or flowmakers used in liquids which could constitute a hazard to health, carry out thorough cleaning and venting of mixer or flowmaker, tank, etc. according to local regulations.

8.1 Contaminated mixer or flowmaker

Warning



If a mixer or flowmaker has been used for a liquid which is injurious to health or toxic, the mixer or flowmaker will be classified as contaminated.

If Grundfos is requested to service the mixer or flowmaker, contact Grundfos with details about the liquid, etc. before you return the mixer or flowmaker for service. Otherwise Grundfos can refuse to accept the mixer or flowmaker for service.

Possible costs of returning the mixer or flowmaker are paid by the customer.

However, any application for service, no matter to whom it may be made, must include details about the liquid if the mixer or flowmaker has been used for liquids which are injurious to health or toxic.

8.2 Service chart

	Type	Service instructions	Lubrication	Inspection
Electric motor	All	Keep the motor housing clean. Otherwise cooling is affected. The motor housing must only be dismantled by Grundfos.	The roller bearings are maintenance-free. They must be replaced if they get noisy.	
Power supply cable	All			Check the power supply cable twice a year for surface damage, strain, kinks, etc. If damaged, the cable must be replaced by Grundfos.
Shaft seal housing	SMD	If they are worn, replace the lip seal and wear ring. If the oil contains water, replace the shaft seal.	If the oil contains water or is contaminated, change the oil. Change the oil at least every two years.	If the shaft seal housing is not monitored for the ingress of water, we recommend that you make inspections every 6 months.
Gearbox	SMG, SFG	If they are worn, replace the lip seal and wear ring. If the oil contains water, replace the shaft seal.	If the oil contains water or is contaminated, change the oil. Change the oil at least every two years. If refilling is required, see section 8.3 Oil .	If you do not monitor the shaft seal housing for the ingress of water, we recommend that you make inspections every 6 months.
Propeller	All			Check the propeller blades regularly for wear and tear. Remove any material wound around the blades, such as ropes, threads, etc. which may cause uneven running and oscillation of the installation. In case of strong turbulence, cleaning is absolutely necessary.
Winch	All	Spray the winch with a protective coating of oil at regular intervals to prevent corrosion.	Lubricate the gear teeth and the bearing bushes twice a year with an all-purpose grease.	
Lifting wire	All	Regular oiling or greasing increases the life of the wire.		Check the wire regularly and always before using the winch. Replace the wire, if required.
Screws	All	Always check that all screws in the motor bracket are properly tightened. Check the screws in the bottom fixation bracket/ bottom fixation plate every time the tank is empty.	When tightening the screws, renew the threadlocker if necessary.	

8.3 Oil

8.3.1 Oil quality, gearbox/shaft seal housing

The gear oil designation for mixers (SMD,SMG) is according to DIN 51502, ISO VG 68. The gear oil designation for flowmakers (SFG) is according to ISO VG 220.

8.3.2 Oil quantity

Type	Gearbox/shaft seal housing [l]
SMD.09 - 18.xx	0.13
SMD.19 - 35.xx	0.26
SMG.09 - 36.xx	1.2
SMG.48 - 120.xx	2.5
SMG.140 - 180.xx	4.0
SFG.xx.130.xx	1.3
SFG.xx.180.xx	3.2
SFG.xx.230.xx	3.2
SFG.xx.260.xx	4.6

8.4 Oil change

Proceed as follows:

1. Place mixer or flowmaker in a horizontal position on supports, and place a pan underneath to collect oil.

- 

Warning

As pressure may have built up in the oil chamber, loosen the oil screw slowly. Do not remove the screw until the pressure has been fully relieved.
2. Remove the screw (2, fig. 44).
 3. Remove the oil drain screw (1), and allow the oil to drain from the chamber into a glass. Leave the oil in the glass for approx. 10 minutes, and check if it contains water. If the oil contains water, replace the shaft seal.

- Note

Dispose of used oil in accordance with local regulations.
4. Clean and refit oil drain screw (1).
 5. Fill oil into the oil chamber through filling hole (2).
The quantity of oil appears from section [8.3.2 Oil quantity](#) and the quality from section [8.3.1 Oil quality, gearbox/shaft seal housing](#).
 6. Refit the screw (2).

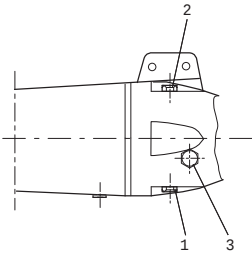


Fig. 44 Position of oil drain and oil filling screws on SMG and SFG

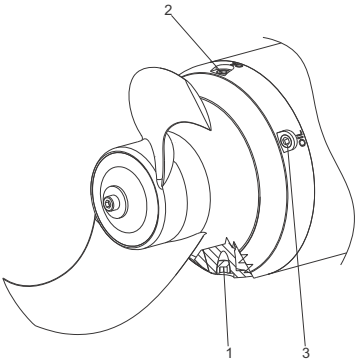


Fig. 45 Position of oil drain and oil filling screws on SMD

9. Fault finding



Warning

Before starting any work on mixers or flowmakers, remove the fuses or switch off the mains switch. Make sure that the power supply cannot be accidentally switched on.

All rotating parts must have stopped moving.

Fault	Cause	Remedy
1. Mixer or flowmaker does not start.	a) No power supply, or supply failure.	Call an electrician.
	b) The power supply cable is faulty.	Call an electrician.
	c) The control system is faulty.	Call an electrician.
	d) The propeller is not free to rotate.	Clean the propeller blades, and check manually that the propeller can rotate freely.
	e) The stator windings are faulty.	Contact Grundfos.
	f) The motor has cut out because of overheating.	Wait until the motor has cooled, and try to restart the mixer or flowmaker.
	g) Different phase voltages.	Call an electrician.
	h) The overload relay is set too low or is faulty.	Check the overload relay. Set the relay to the rated current. See section 5.3 Overload relays .
	i) The leakage sensor has cut out mixer or flowmaker.	Contact Grundfos.
	j) Humidity in motor.	Contact Grundfos.
2. Mixer or flowmaker starts but stops immediately.	a) The stator windings are faulty.	Contact Grundfos.
	b) Different phase voltages.	Call an electrician.
	c) The overload relay is set too low or is faulty.	Check the overload relay. Set the relay to the rated current. See section 5.3 Overload relays .
	d) The leakage sensor has cut out mixer or flowmaker.	Contact Grundfos.
	e) Humidity in motor.	Contact Grundfos.
3. No or inadequate circulation produced even if the motor is running.	a) The propeller rotates in the wrong direction.	Interchange two phases of the mains supply.
	b) The mixer or flowmaker runs on two phases.	• Check electrical connections. • Replace faulty fuses. • Call an electrician.
	c) The internal parts are worn.	Contact Grundfos.
	d) The propeller blades are dirty or damaged.	Clean the blades, and inspect for any wear. If the propeller blades are worn or damaged, contact Grundfos.
4. Mixer or flowmaker runs unevenly and is noisy.	a) The internal parts are worn.	Contact Grundfos.
	b) The propeller blades are dirty or damaged.	Clean the blades, and inspect for any wear. If the propeller blades are worn or damaged, contact Grundfos.
	c) Faulty motor or gearbox roller bearings.	Contact Grundfos.
	d) Oscillations caused by the installation, resonance.	Check the installation design.

Fault	Cause	Remedy
5. High current and power consumption.	a) Wrong voltage supply or supply failure.	Call an electrician.
	b) The power supply cable is faulty.	Call an electrician.
	c) The control system is faulty.	Call an electrician.
	d) The propeller is not free to rotate.	Clean the propeller blades, and check manually that the propeller can rotate freely.
	e) The stator windings are faulty.	Contact Grundfos.
	f) The mixer or flowmaker runs on two phases.	• Check electrical connections. • Replace faulty fuses. • Call an electrician.
	g) The internal parts are worn.	Contact Grundfos.
	h) Faulty motor or gearbox roller bearings.	Contact Grundfos.

10. Technical data

10.1 General technical data

Voltage tolerance		
	400-415 V	- 10 %/+ 6 %
	380 V	- 5 %/+ 10 %
	440-480 V	- 10 %/+ 10 %
Enclosure class	IP68	
Insulation class		
	SMG, SFG	H
	SMD.19 - 35	H
	SMD.09 - 18	F
Maximum installation depth	20 m below liquid surface	
Maximum number of starts per hour		
	SMG, SFG	20
	SMD	60
Power supply cable length	10 m, 15 m*	
Wire length on all winches	10 m (standard)	

* The standard cable length is 10 and 15 m. Other cable lengths are available on request.

The mixers and flowmakers are designed for continuous operation.

10.2 Motor

Seal	Mechanical shaft seal	
Material, motor housing		
SMG, SFG	Cast iron, grade 25 (EN-GJL-250)	
SMD	Cast stainless steel 1.4408	

10.3 Gearbox, SMG, SFG only

Type	Planetary gearbox	
Gears	Hardened and ground steel	
Seal monitoring	Leakage sensor incorporated in gearbox	
Drive-end bearings	Two tapered roller bearings	
Material, gear casing	Cast iron, grade 25 (EN-GJL-250)	

10.4 Shaft seals

Sealing against ingress of surrounding liquid

Two lip seals and one mechanical shaft seal made of tungsten carbide/tungsten carbide or SiC/SiC

10.5 Propeller

SMD

Number of blades	2
Propeller diameter	180-370 mm
Material, propeller with hub	Stainless steel

SMG

Number of blades	2
Propeller diameter	400-1000 mm
Material, propeller with hub	Stainless steel

SFG.xx.130

Number of blades	2
Propeller diameter	1300 mm
Material, propeller with hub	Polyurethane (Baydur®) with stainless-steel hub

SFG.xx.150-260.xx

Number of blades	2 or 3
Propeller diameter	1500, 1800, 2300, 2600 mm
Material, hub	Cast iron (EN-GJS-400-15)
Material, propeller blades	Polyurethane (Baydur®) with cast-iron reinforcement

10.6 Sound pressure level

The sound pressure level of the mixer or flowmaker is lower than 70 dB(A).

11. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

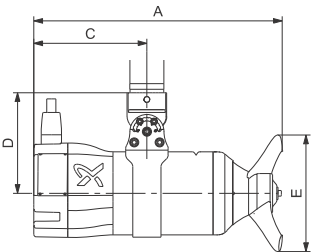
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Appendix

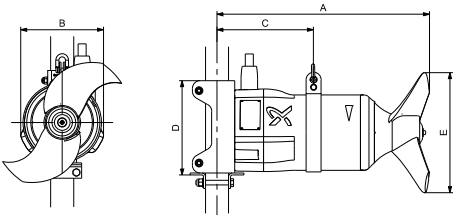
DIMENSIONS AND WEIGHTS

SMD

SMD.10.18.1765.T.
SMD.13.21.1750.T.
SMD.17.25.1725.T.



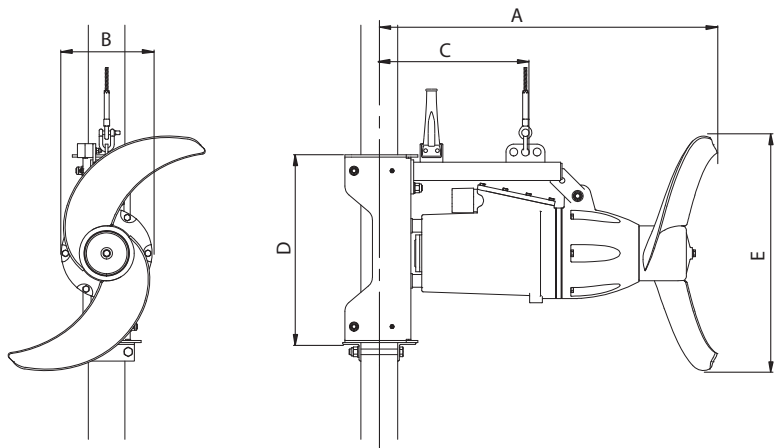
SMD.10.18.1765.
SMD.13.21.1750.
SMD.17.25.1725.
SMD.22.29.1174.
SMD.26.32.1165.
SMD.32.37.1152.



Type	Rated power [kW]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	Net weight* [kg]
SMD.10.18.1765.T.	1.0	440	180	195	180	180	36
SMD.13.21.1750.T.	1.3	440	180	195	180	210	36
SMD.17.25.1725.T.	1.7	440	180	195	180	250	36
SMD.10.18.1765.	1.0	490	180	225	240	180	38
SMD.13.21.1750.	1.3	490	180	225	240	210	38
SMD.17.25.1725.	1.7	490	180	225	240	250	38
SMD.22.29.1174.	2.2	550	230	240	240	290	67
SMD.26.32.1165.	2.6	550	230	240	240	320	67
SMD.32.37.1152.	3.2	550	230	240	240	370	67

* With motor bracket and 10 m cable. Weight of cable: 0.5 kg/m.

TM06 5319 4315 - TM06 5320 4315



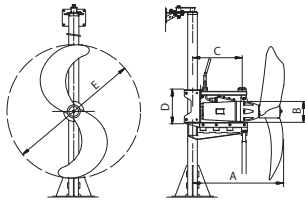
TN02 4944 32 10

Type	Rated power [kW]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	Net weight* [kg]
SMG.09.55...	0.9	780	200	320	410	550	79
SMG.12.63...	1.2	790	200	320	410	630	79
SMG.16.63...	1.6	790	200	320	410	630	79
SMG.20.71...	2.0	800	200	320	410	710	86
SMG.25.71...	2.5	800	200	320	410	710	86
SMG.33.71...	3.3	800	200	320	410	710	86
SMG.38.71...	3.8	800	200	320	410	710	86
SMG.56.86...	5.6	1050	260	430	460	860	174
SMG.70.86...	7.0	1050	260	430	460	860	174
SMG.95.86...	9.5	1050	260	430	460	860	190
SMG.120.86...	12.0	1050	260	430	460	860	190
SMG.160.90...	16.0	1100	315	460	460	900	280

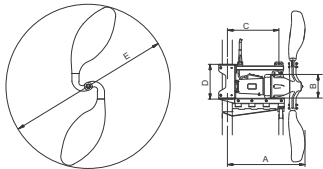
* With motor bracket and 10 m cable. Weight of cable: 0.5 kg/m.

SFG.xx.130-230.xx

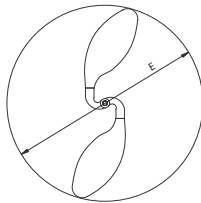
SFG.xx.130.xx



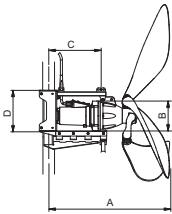
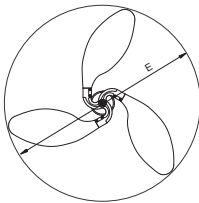
SFG.07-26.180.xx



SFG.07-22.230

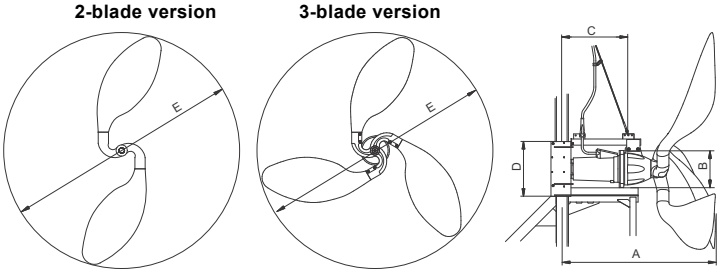


SFG.32.180
SFG.36.180
SFG.26-36.230



Type	Propeller version	Rated power [kW]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	Net weight [kg]
SFG.07.130...	2-blade	0.7	900	200	353	400	1300	113
SFG.10.130...		1.0						
SFG.14.130...		1.4						
SFG.17.130...		1.7						
SFG.22.130...		2.2						118
SFG.29.130...		2.9						
SFG.33.130...		3.3						
SFG.36.130...		3.6						
SFG.07.180...	3-blade	0.7	1200	302	570	400	1800	195
SFG.10.180...		1.0						
SFG.14.180...		1.4						
SFG.17.180...		1.7						
SFG.22.180...		2.2						200
SFG.29.180...		2.9						
SFG.35.180...		3.5						
SFG.38.180...		3.8						
SFG.07.230...	2-blade	0.7	1200	302	570	400	2300	200
SFG.09.230...		0.9						
SFG.12.230...		1.2						
SFG.16.230...		1.6						
SFG.19.230...		1.9						205
SFG.22.230...		2.2						
SFG.25.230...	3-blade	2.5						230
SFG.32.230...		3.2						
SFG.38.230...		3.8						

TM02 6346 0305 - TM02 6345 0305 - TM04 3959 0409



TM04 3957 0409

Type	Propeller version	Rated power [kW]	A [mm]	B [mm]	C [mm]	D [mm]	E [mm]	Weight incl. motor bracket and 10 m cable [kg]
SFG.22.260.29...	2-blade	2.2	1500	360	650	585	2600	360
SFG.32.260.34...		3.2						
SFG.36.260.35...		3.6						
SFG.44.260.38...		4.4						
SFG.50.260.35...	3-blade	5.0	1500	360	680	585	2660	415
SFG.60.260.38...		6.0						
SFG.72.260.40...		7.2						
SFG.80.260.42...		8.0						

Declaration of conformity

GB: EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products SMD, SMG, SFG, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EU member states.

AR: إقرار مطابقة EU

نقرر نحن، جرونډفوس، بمقتضى مسؤوليتنا الفردية بأن المنتجين SMD، SMG، SFG، اللذين يختص بهما الإقرار أدناه، يكونان مطابقيين لتوجيهات المجلس المذكورة أدناه بشأن التقريب بين قوانين الدول أعضاء المجموعة الأوروبية/الاتحاد الأوروبي (EU).

PT: Declaração de conformidade UE

A Grundfos declara sob sua única responsabilidade que os produtos SMD, SMG, SFG, aos quais diz respeito a declaração abaixo, estão em conformidade com as Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da UE.

KO: EU 적합성 선언

Grundfos 는 아래의 선언과 관련된 SMD, SMG, SFG 제품이 EU 회원국 법률에 기하여 아래의 이사회 지침을 준수함을 단독 책임 하에 선언합니다.

CN: 欧盟符合性声明

我们，格兰富，在我们的全权责任下声明，产品 SMD, SMG、SFG，即该合格证所指之产品，欧盟使其成员国法律趋于一致的以下理事会指令。

- Machinery Directive (2006/42/EC)
Standard used: EN ISO 12100: 2010
- Low Voltage Directive (2014/35/EU)
Standards used: EN 60034 and EN 60204:2006 + A1:2009
- EMC Directive (2014/30/EU)
Standards used: EN 61000-6-2:2005 and EN 61000-6-3:2007 + A1:2011

This EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 98826143 0616)

Schachen, 1st of April 2016



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